A classic example of the impact of ecology is the growth and decline of IBM. In the 1950s IBM was an important but struggling company, with a good punch-card business and a good typewriter business. Then the computer came along. Why IBM took over that market, and Burroughs, General Electric, Univac, RCA, Honeywell, and Control Data ended up losers, is a complicated story, but the main element, I believe, was IBM's marketing strategy. Its hardware was generally satisfactory, but not great; its marketing, however, was tremendous. It sold its mainframes because it controlled the executive suites of the country. Time after time I heard stories about corporate computer departments recommending a Honeywell or Burroughs computer, but when the purchase order reached the president's desk, he crossed out "Burroughs" and wrote in "IBM." It wasn't because he thought IBM had a better computer for the money, but he liked and trusted the IBM salesman. That, of course, generated profits for IBM that it could then put back into new equipment and programs, until it dominated the market, the way NCR had done before it and Microsoft did after it.

It seemed as if IBM had an unbreakable hold on the computer market—until there was a dramatic shift in the ecology. In the same way that a huge meteorite hit the earth and changed the climate and wiped out the dinosaurs, the invention of the personal computer broke IBM's domination. The dinosaurs couldn't adapt. IBM tried to adapt but was late about it. When it finally got around to making a PC, the product was sufficiently competitive, but IBM's big advantage—control of the purchasing decision—had disappeared.

By that time the buy decision at companies had been pushed down to a different level, because the purchase involved a $2,000 machine instead of a $5 million machine. Department heads are perfectly competent to decide where to spend $10,000 on hardware. Who cares? The marketing decision no longer went up to the president's desk. IBM's selling power was suddenly cut off; the company had suffered a marketing aneurysm. Deals now had to be made with the wrong person (from IBM's point of view), because a